

Question 1.2. State the null hypothesis. It should reflect the position of either Dr. DeNero or Dr. Sahai. (4 points)

Note: Check out [11.3](#) for a refresher on hypotheses.

The probability of getting vaccine 1 is $3/5$, and the probability of getting vaccine 2 is $2/5$, reflecting Dr. DeNero's position.

Question 1.3. State the alternative hypothesis. It should reflect the position of the doctor you did not choose to represent in Question 1.2. **(4 points)**

Note: Check out [11.3](#) for a refresher on hypotheses.

The probability of getting vaccine 1 is not $3/5$, and the probability of getting vaccine 2 is not $2/5$, reflecting Dr. Sahai's position.

Question 1.7. Simulate 20,000 values of the test statistic under the assumption you picked in Question 1.6. (4 points)

As usual, start by defining a function that simulates one value of the statistic. Your function should use `sample_proportions`. (You may find a variable defined in Question 1.1 useful here!) Then, write a `for` loop to simulate multiple values and collect them in the array `simulated_statistics`.

Use as many lines of code as you need. We have included the code that visualizes the distribution of the simulated values. The red dot represents the observed statistic you found in Question 1.5.

```
In [11]: def one_simulated_statistic():
         values = sample_proportions(318, [3/5,2/5]).item(0)
         return abs(values-0.6)*100
```

```
In [12]: # Run the this cell a few times to see how the simulated statistic changes
         one_simulated_statistic()
```

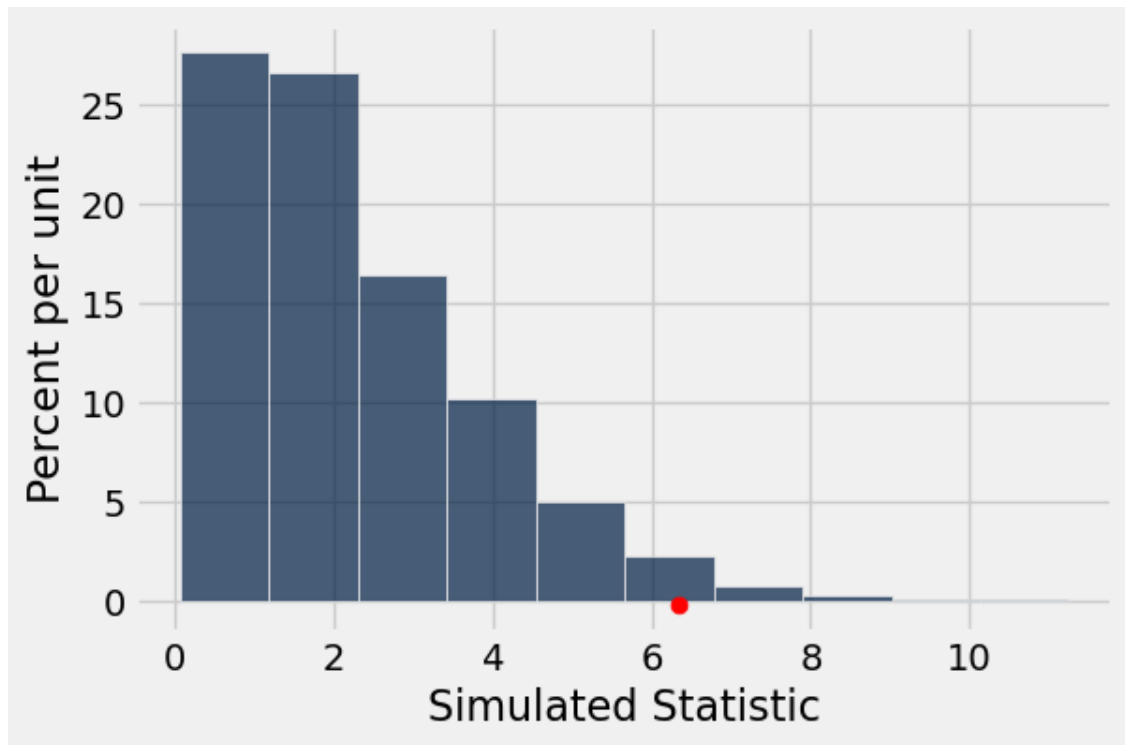
```
Out[12]: 1.823899371069182
```

```
In [13]: num_simulations = 20000

         simulated_statistics = make_array()
         for i in np.arange(num_simulations):
             simulated_statistics = np.append(simulated_statistics, one_simulated_statistic())
```

```
In [14]: # Run this cell to produce a histogram of the simulated statistics

         Table().with_columns('Simulated Statistic', simulated_statistics).hist()
         plt.scatter(observed_statistic, -0.002, color='red', s=40);
```



Question 2.1. Suppose we want to test whether or not each factor contributes the same amount to the overall Happiness Score. Define the null hypothesis, alternative hypothesis, and test statistic in the cell below. **(4 points)**

Note: Please format your answer as follows: - Null Hypothesis: ...

- Alternative Hypothesis: ...

- Test Statistic: ...

Null Hypothesis: Each happiness factor contributes $1/6$ of the overall Happiness Score.

Alternative Hypothesis: Each happiness factor does not contribute $1/6$ of the overall Happiness Score.

Test Statistic: The total variation distance between the distributions of each happiness factor.

Question 2.5. What can you conclude about how each factor contributes to the overall happiness score in the US? Explain your answer using the results of your hypothesis test. Assume a p-value cutoff of 5%. (4 points)

Since the p-value is less than the cutoff of 0.05, the null hypothesis is rejected. So, our evidence suggests that each happiness factor does not contribute 1/6 of the overall Happiness Score.

